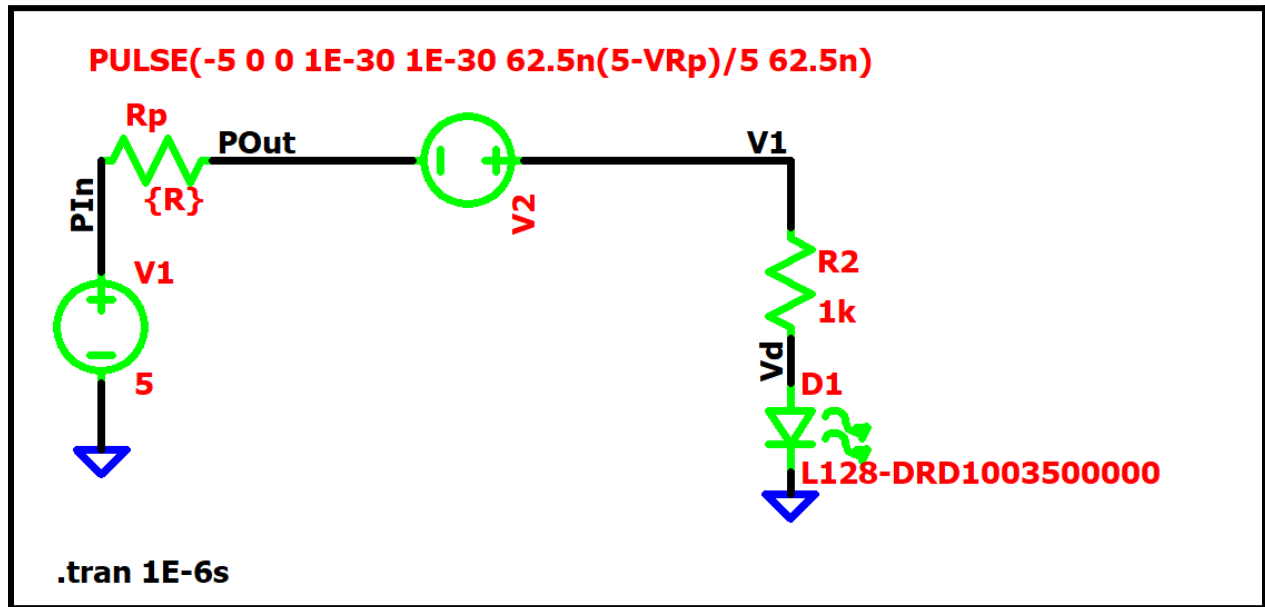


Pulse-Width Modulation (PWM) Control of an LED

Circuit Schematic:



Although we can't run this simulation in this state as LTSpice doesn't allow the T_{on} to be a variable, this best represents our schematic. We used the standard Arduino frequency of 16 MHz, which produces a period of 62.5 ns. The circuit will adjust its T_{on} , and therefore its duty cycle, based on the input voltage, P_{Out} , which will depend on the resistance across the potentiometer; this equation is $T_{on} = 62.5ns(V_{P_{Out}}/5) = 62.5ns((5 - V_{Rp})/5)$. The initial voltage and maximum voltage will be shifted due to the way Arduino outputs its voltage independently from the input, unlike LTSpice. The voltages will be -5 and 0 in order to best represent the shift.

Description:

We will be proving the concept of using a voltage divider containing a potentiometer to control the output of a microcontroller, which is a resistor in series with an LED. Using the potentiometer, we will prove that the brightness can be adjusted. The potentiometer in the voltage divider will control the amount of voltage being input, P_{Out} , into the Arduino as the resistance of the potentiometer, R_p , increases, the input voltage decreases, and vice versa. We

will code the Arduino to use this input in a formula that controls the duty cycle of the PWM signals. This will change the brightness of the LED by changing the average voltage the Arduino is outputting. There will also be a current-limiting resistor, R_2 , in series with the LED to ensure there is no damage done to the LED.

Analysis:

Use the formula for a voltage divider to calculate the voltage of R resulting from the potentiometer at a given resistance. We used a red LED meaning the forward voltage of the LED will be close to 1.8 V, meaning this will be the voltage across the LED. To demonstrate the current, we took measurements at 3 resistances of $5k\Omega$, $0k$, and $2.61k\Omega$.

Voltage divider formula: $V_{Rp} = V_s \frac{R_2}{R_1 + R_2}$

Where V_s is the voltage source of 5V,

R_1 is the potentiometer resistance closest to the input (0 to R_1),

And R_2 is the potentiometer resistance closest to ground (total - R_1)

To find the resulting current through the resistor and LED, use Ohm's law: $V = IR$

Where V is the Vout of the potentiometer

And R is a $1k\Omega$ resistor

When the potentiometer has resistance $5k\Omega$,

Plugging into the voltage divider formula, we get: $V_{Rp} = 5 \frac{5k}{5k + (5k - 5k)} = 5(1) = 5V$

$T_{on} = 62.5ns((5-5)/5) = 0$ ns so the duty cycle will equal

$(0/62.5ns) * 100\% = 0\%$

To find the current:

$$I = \frac{V}{R} = \frac{0}{1k} = 0 \text{ A or } 0mA$$

For when the potentiometer has a resistance of $2.61k\Omega$,

$$\text{By plugging into the formula: } V_{Rp} = 5 \frac{2.61k}{2.61k + (5k - 2.61k)} = 5 \left(\frac{2.61k}{5k} \right) = 2.61V$$

$T_{on} = 62.5ns \left(\frac{5 - 2.61}{5} \right) = 29.875ns$ so the duty cycle will equal

$$(29.875ns / 62.5ns) * 100\% = 47.8\%$$

To find the current:

$$I = \frac{V}{R} = \frac{2.39 - 1.8}{1k} = \frac{0.59}{1k} = 0.00059 A \text{ or } 0.59 mA$$

And for when the potentiometer has a resistance of $0k\Omega$,

$$\text{Plugging in we get, } V_{Rp} = 5 \frac{0k}{0 + (5k - 0k)} = 5(0) = 0V$$

$T_{on} = 62.5ns \left(\frac{5 - 0}{5} \right) = 62.5ns$ so the duty cycle will equal

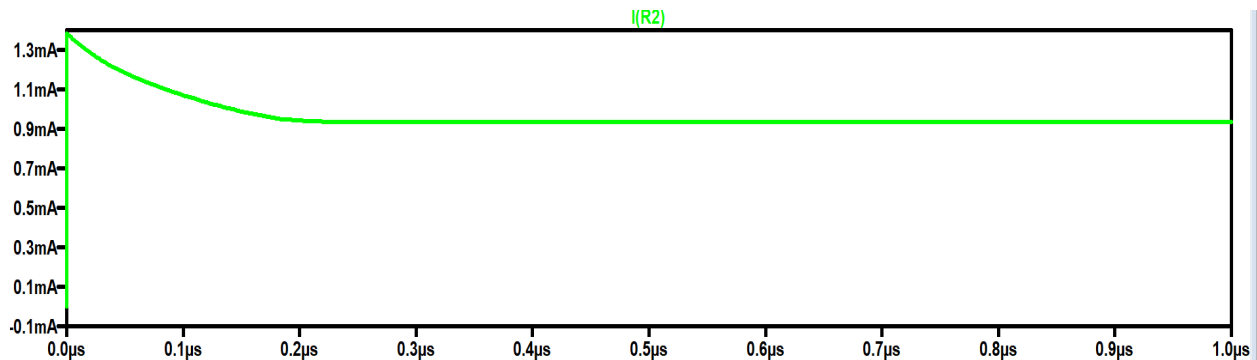
$$(62.5ns / 62.5ns) * 100\% = 100\%$$

To find the current:

$$I = \frac{V}{R} = \frac{5 - 1.8}{1k} = \frac{3.2}{1k} = 0.0032 A \text{ or } 3.2 mA$$

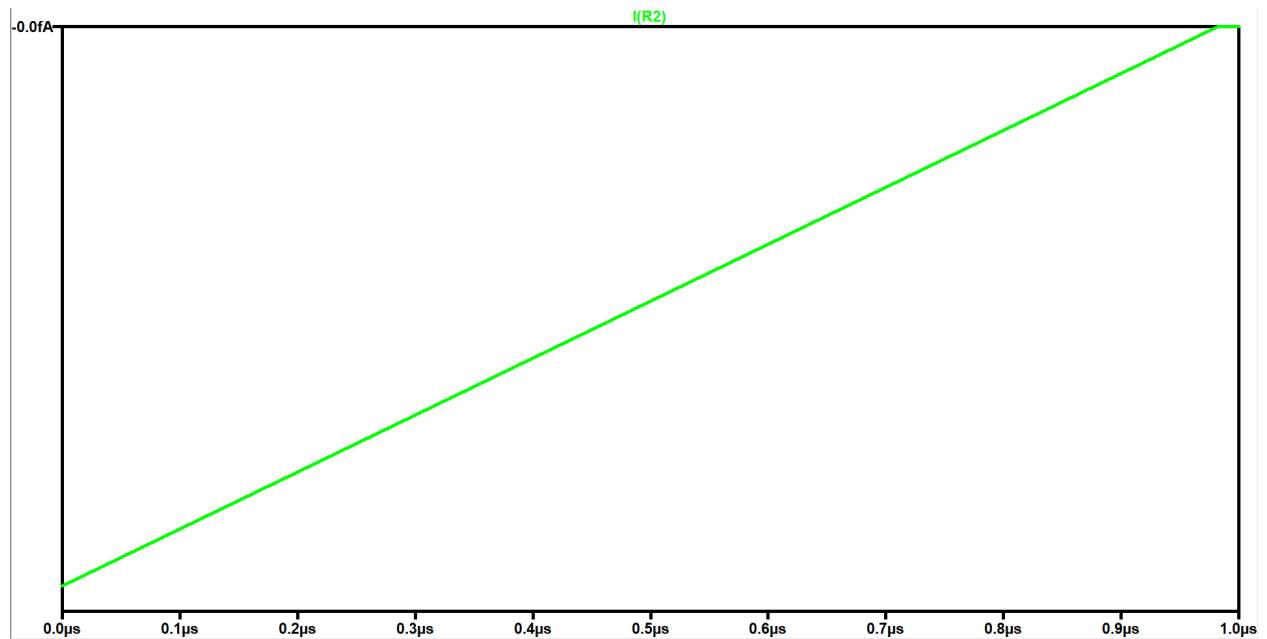
Simulation:

For when potentiometer is $2.61k\Omega$:



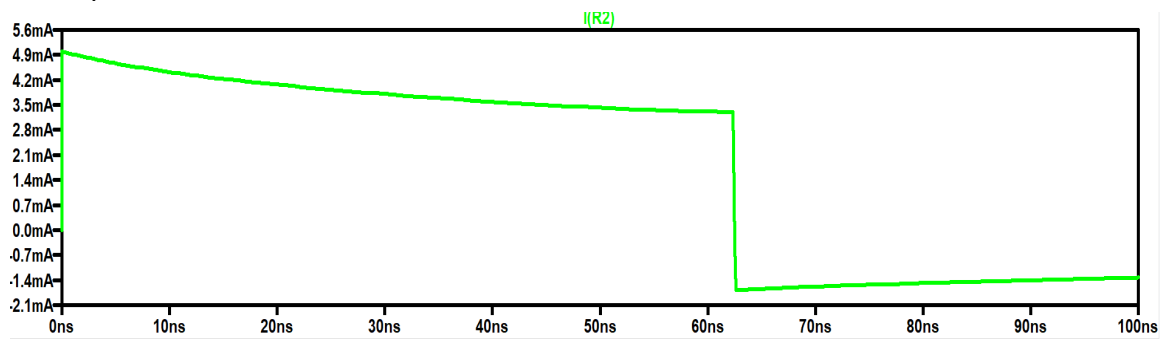
Average current of R_2 is around 0.95mA

When potentiometer is $5k\Omega$:



Average current of R_2 is 0 mA .

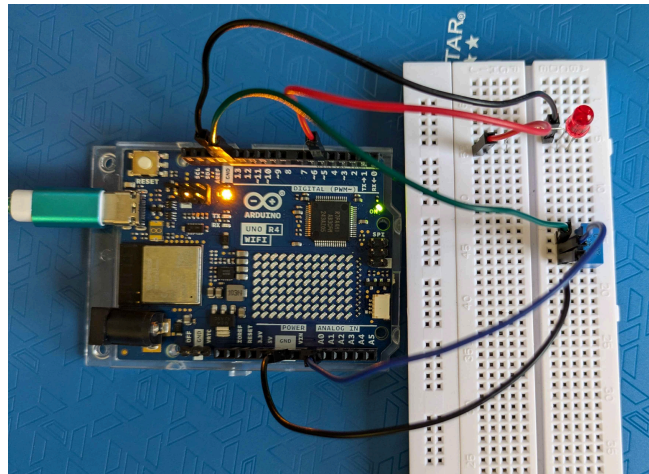
When potentiometer is $0k\Omega$:



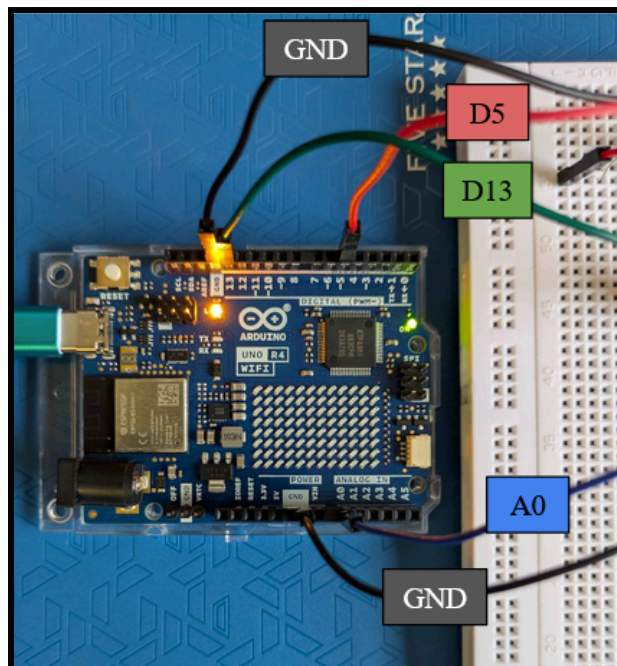
Average current of R_2 for one cycle is about 3.7 mA .

Measurement:

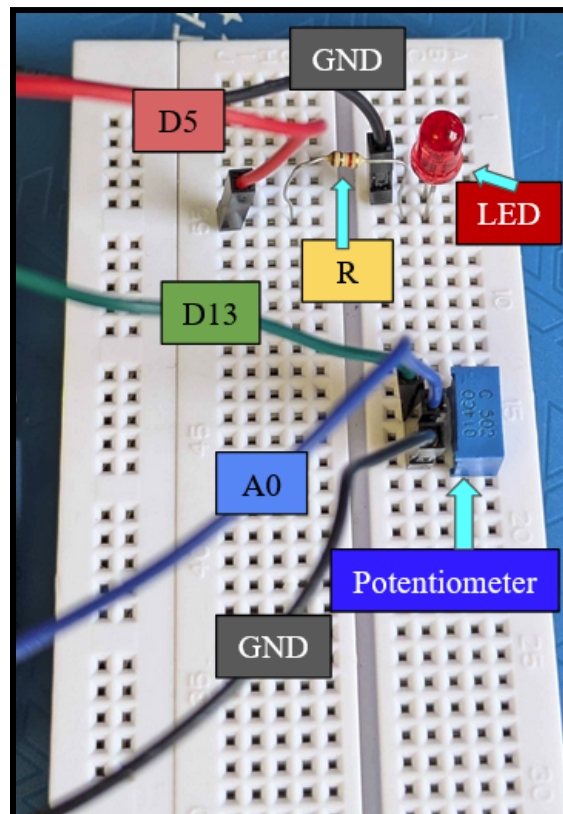
Full diagram of circuit:



D describes pins, ex: D5 is pin 5

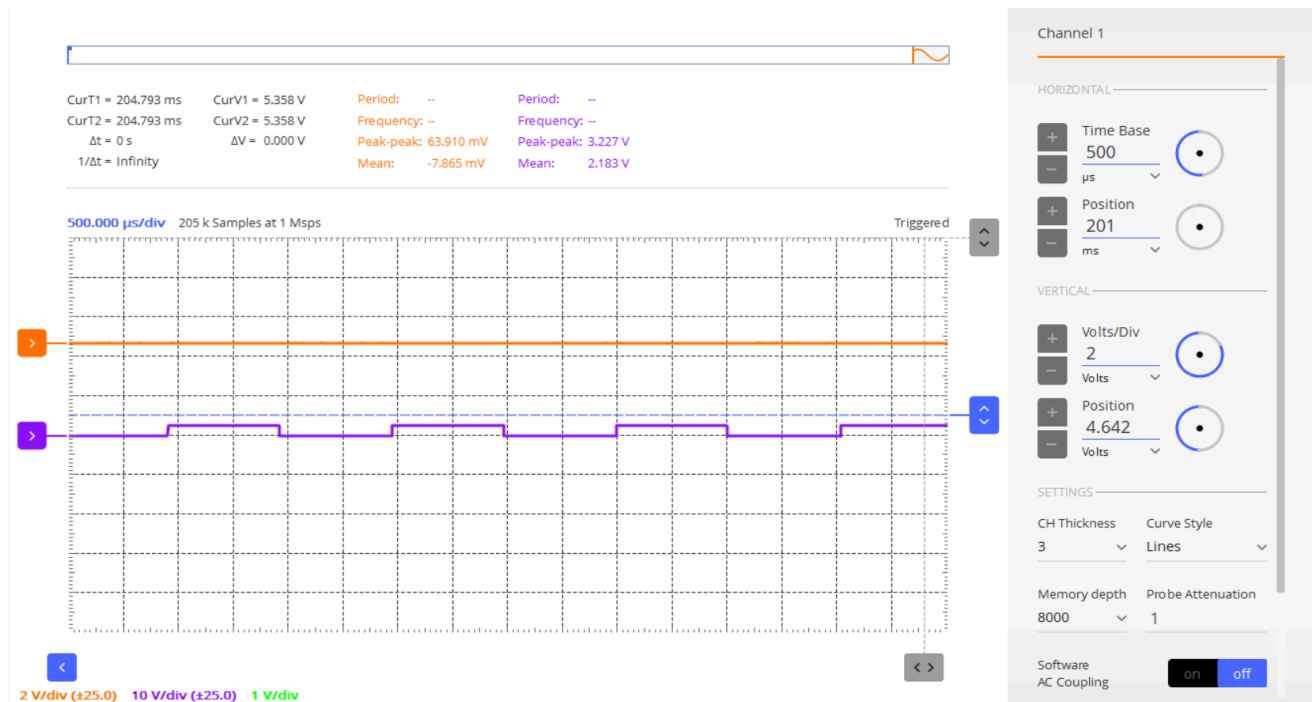
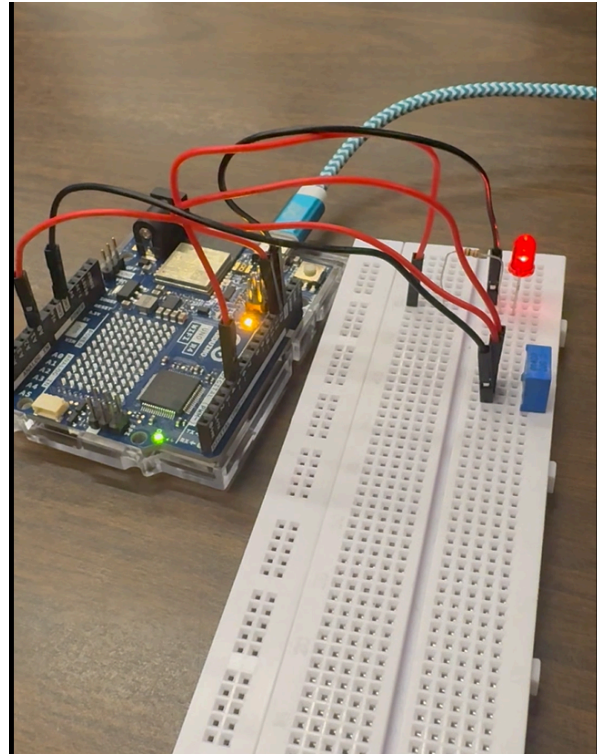
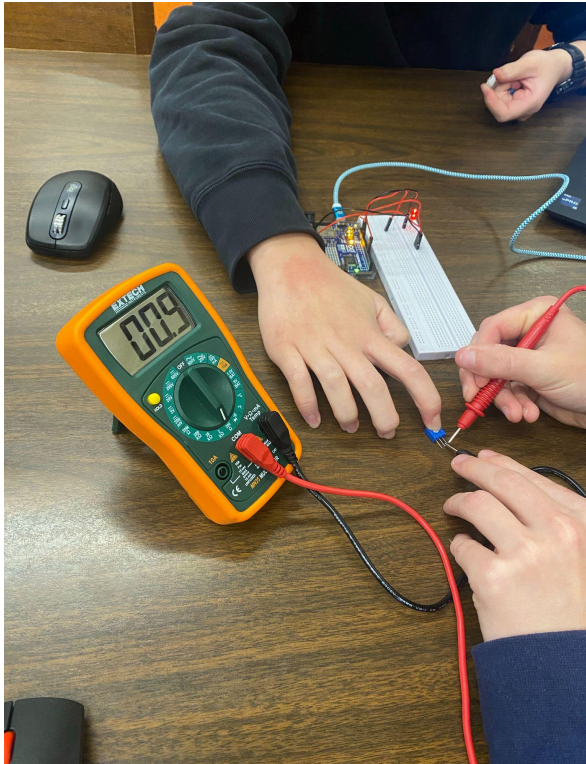


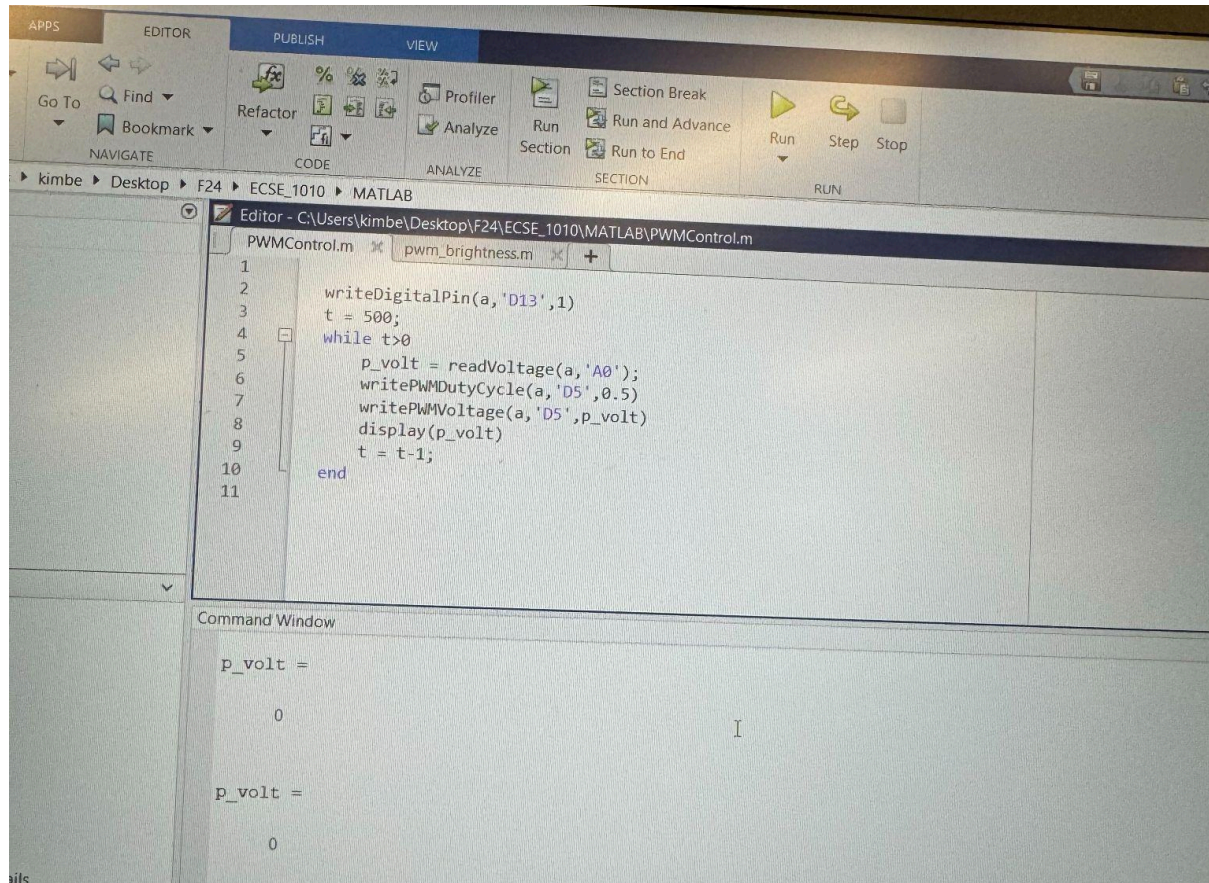
$R = 1k\Omega$



Potentiometer: max resistance of $5k\Omega$

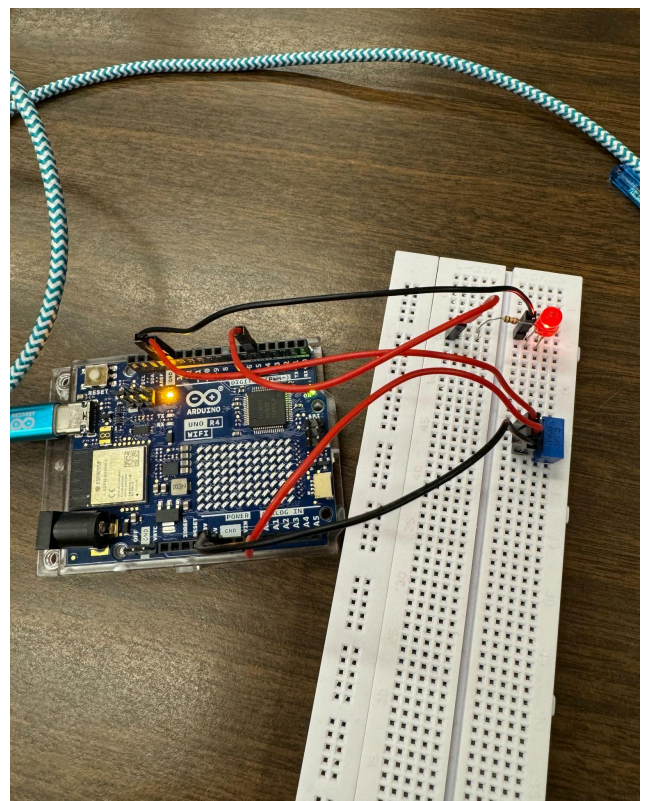
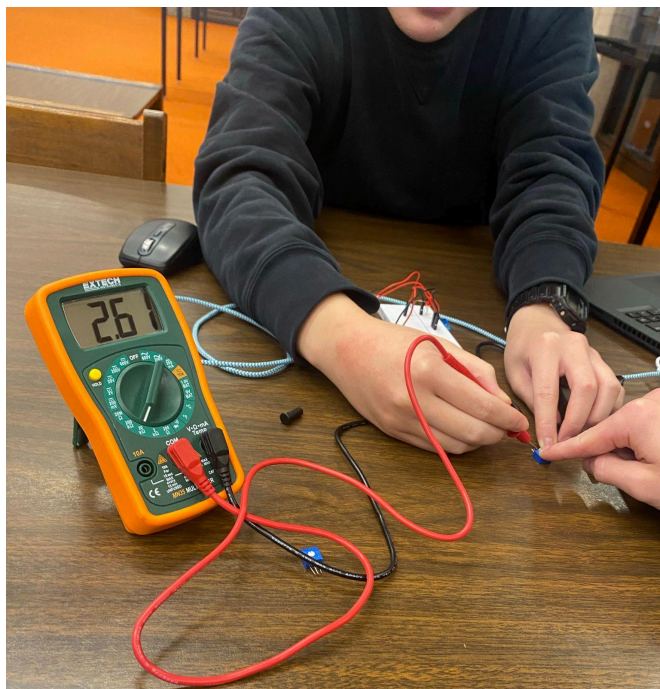
0Ω

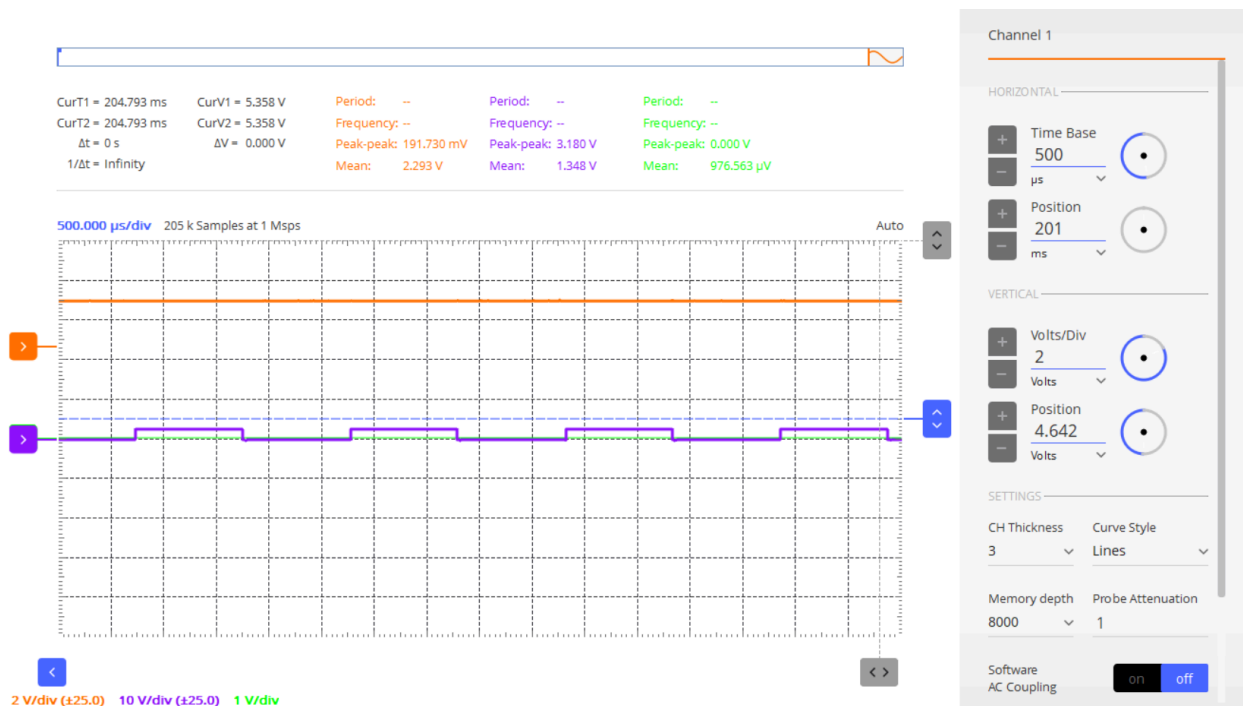
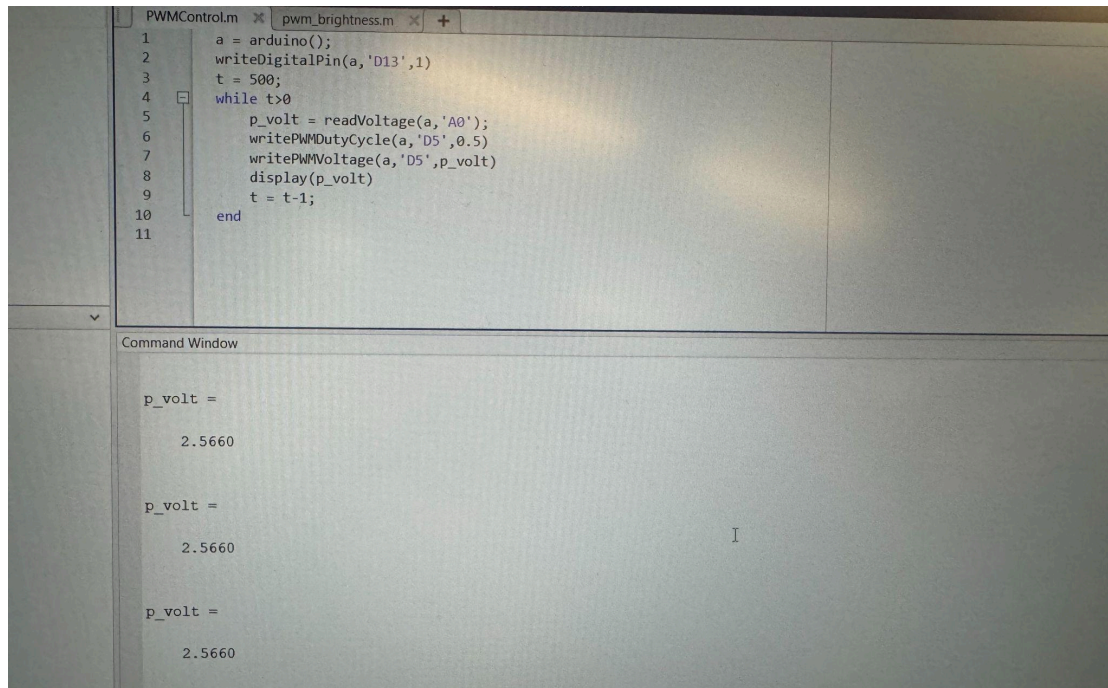




When the potentiometer has 0 resistance, the voltage is measured at -7.9 mV, which is basically 0, and the voltage across the current limiting resistor is 2.183 V, indicating the current across it and the LED is $i = 2.183 \text{ V} / 1\text{k} = 2.183 \text{ mA}$. As you can see, the LED has a constant, bright red appearance.

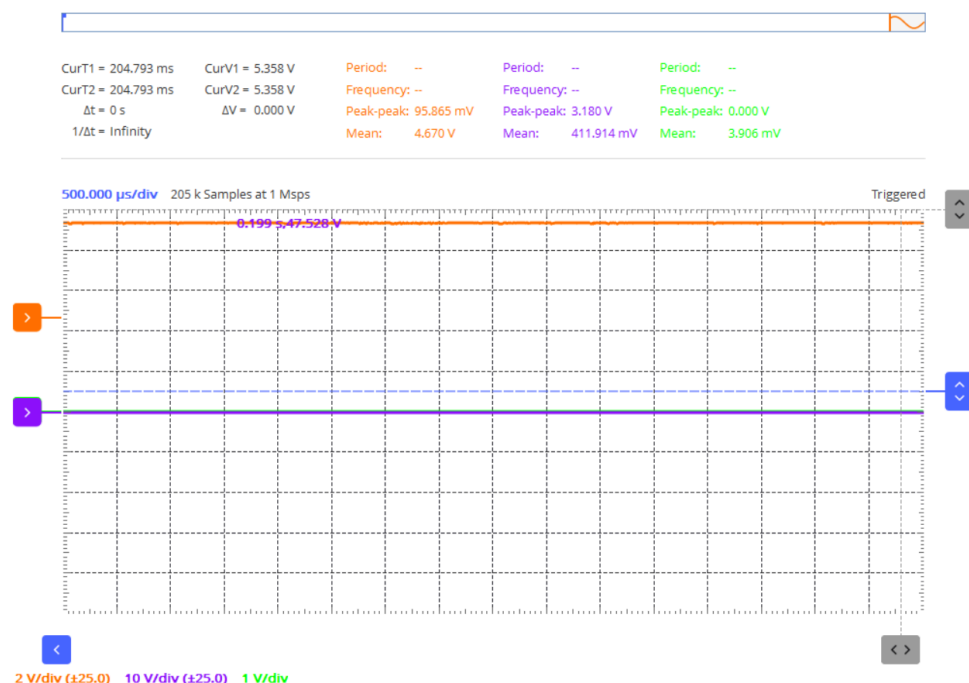
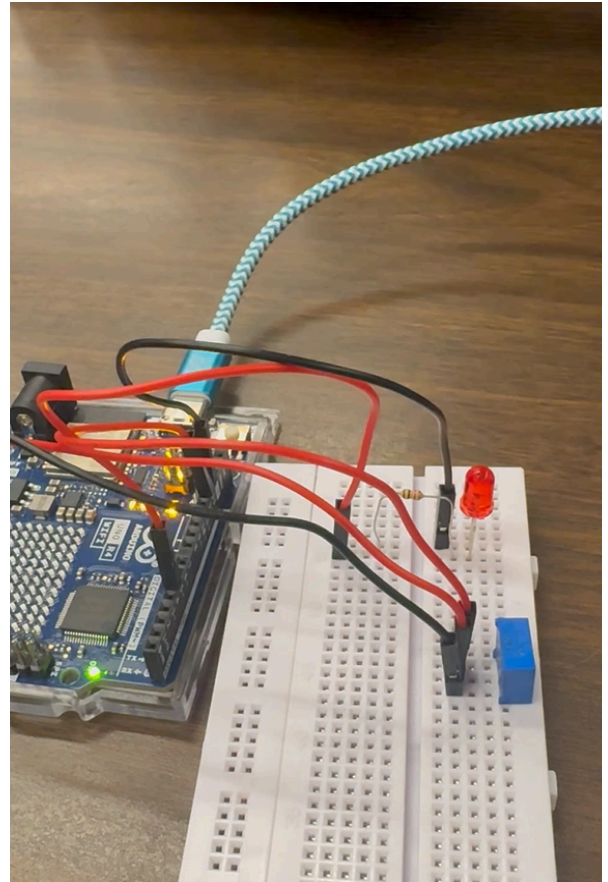
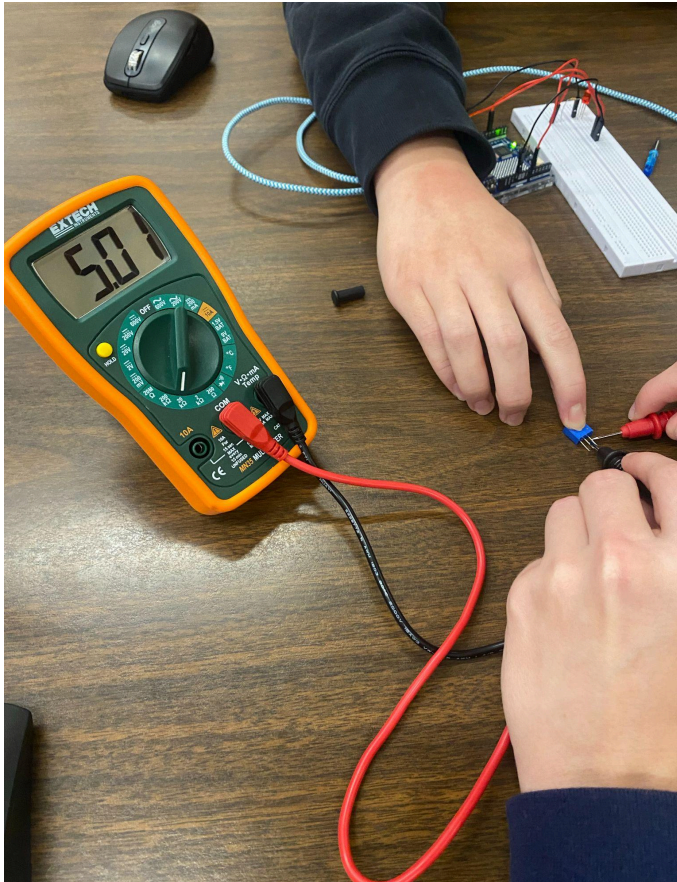
2.61k Ω

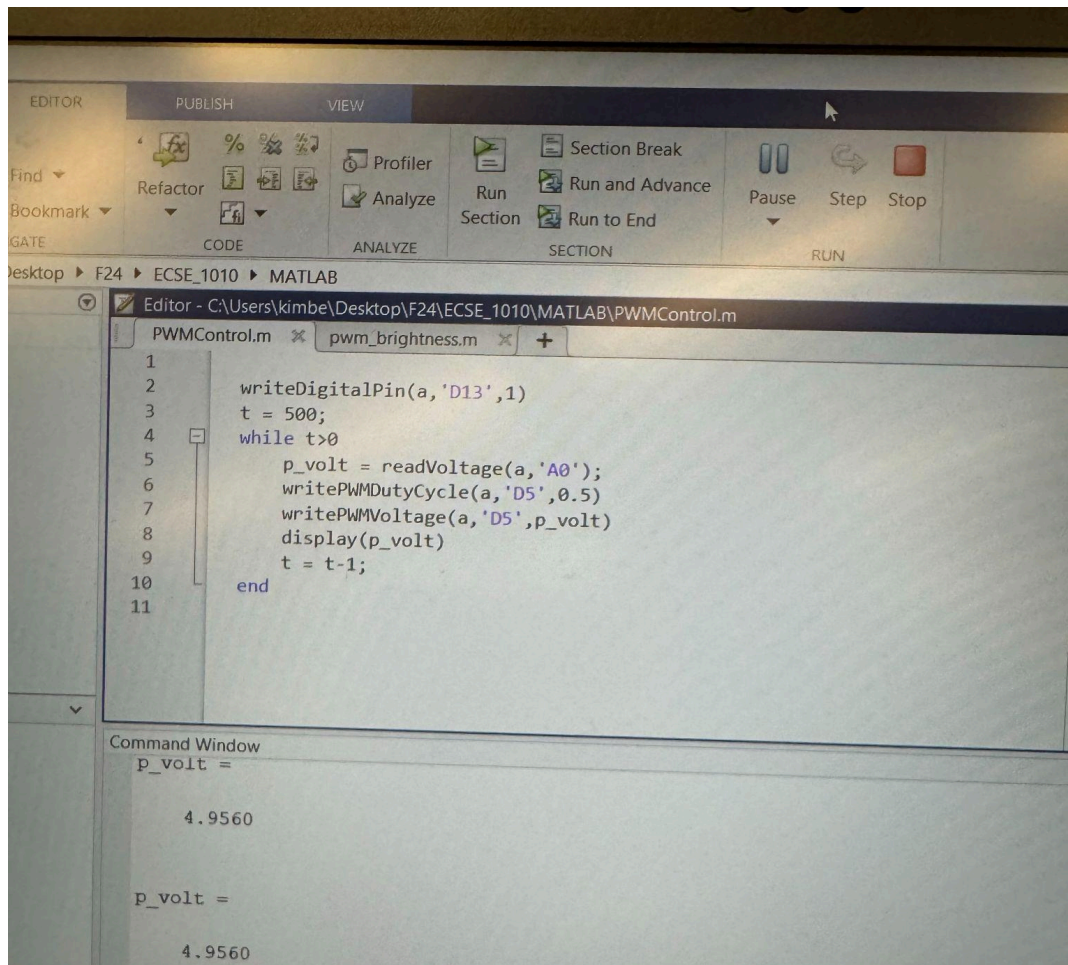




When the potentiometer has $2.61\text{k}\Omega$ resistance, the voltage is measured at 2.29 V and the voltage across the current limiting resistor is 1.348 V, indicating the current across it and the LED is $i = 1.348\text{ V} / 1\text{k} = 1.348\text{ mA}$. As you can see, the LED has a constant, red appearance with medium brightness.

5k Ω





The screenshot shows the MATLAB IDE interface. The top menu bar includes EDITOR, PUBLISH, and VIEW. The toolbar contains various icons for editing, running, and debugging. The main editor window displays the file PWMControl.m with the following code:

```
1 writeDigitalPin(a,'D13',1)
2 t = 500;
3 while t>0
4     p_volt = readVoltage(a,'A0');
5     writePWMDutyCycle(a,'D5',0.5)
6     writePWMMVoltage(a,'D5',p_volt)
7     display(p_volt)
8     t = t-1;
9 end
10
11
```

The Command Window at the bottom shows the output of the script:

```
Command Window
p_volt =
    4.9560

p_volt =
    4.9560
```

When the potentiometer has $5k\ \Omega$ resistance, the voltage is measured at 4.67 V, and the voltage across the current limiting resistor is 412 mV, indicating the current across it and the LED is $i = 412\text{ mV} / 1k = 0.412\text{ mA}$. As you can see, the LED has low brightness.

Discussion (and answer related questions in Alpha Lab):

When $R_{pot} = 5k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	5	5	4.67	6.60%
Current through R and LED(mA)	0	0	0.412	N/A

When $R_{pot} = 0k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	0	0	-0.007	N/A
Current through R and LED(mA)	3.2	2.6	2.183	16.04%

When $R_{pot} = 2.61k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	2.61	2.61	2.29	12.26%
Current through R and LED (mA)	0.59	0.95	1.348	15.71%

From our results for each value of resistance of the potentiometer, while there is a considerable percentage of error, it isn't large enough to say that our values from analysis, simulation, and measurement aren't consistent at all. The largest percent errors are in the current values, which we chose to calculate in units of mA instead of A. Due to this unit difference, the percent error increases as the value of the measurements excusing the units is larger than they would be as just amps. Some values we simply couldn't calculate a percent error for since they were being compared to 0, which results in a calculation error as you cannot divide by 0. Another small but possible reason for why our values don't match up exactly are the values we use in simulation

and analysis are theoretical values, and specifically for our simulation on LT Spice, the values had to be rounded, and we couldn't account for the correct values of the LED.